Double Integral Revision

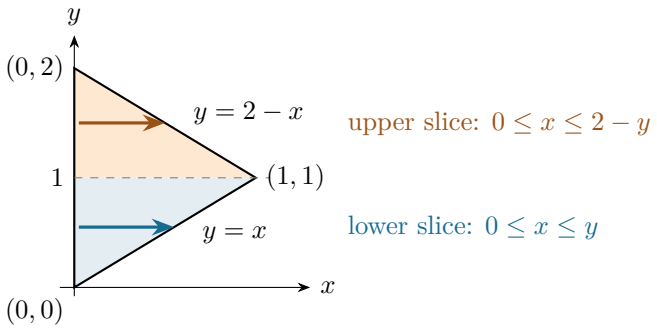
AS1109 Quiz 2: 10 marks. Section H: 26 October, 4:00–4:50 pm, EB1-103.

For a plane region R, $A = \iint_R 1\,\mathrm{d}A$. For a nonnegative height h, $V = \iint_R h\,\mathrm{d}A$. For a solid between two surfaces, $h = z_{\text{top}} - z_{\text{bottom}}$. With $\mathrm{d}y\,\mathrm{d}x$, use vertical slices; with $\mathrm{d}x\,\mathrm{d}y$, use horizontal slices.

1. Reverse the order when one boundary changes

$$I = \int_0^1 \int_x^{2-x} f(x,y)\,\mathrm{d}y\,\mathrm{d}x.$$

The bounds describe $0 \le x \le 1$ and $x \le y \le 2 - x$. The triangle has vertices $(0,0)$, $(0,2)$ and $(1,1)$; the sloping lines meet because $x = 2 - x$ gives $x = y = 1$.



The full new outer range is $0 \le y \le 2$. A horizontal slice always starts at $x = 0$, but its endpoint changes:

$$\begin{array}{l|l} 0 \le y \le 1 & 0 \le x \le y \\ 1 \le y \le 2 & 0 \le x \le 2 - y \end{array}$$

Thus

$$\boxed{I = \int_0^1 \int_0^y f(x,y)\,\mathrm{d}x\,\mathrm{d}y + \int_1^2 \int_0^{2-y} f(x,y)\,\mathrm{d}x\,\mathrm{d}y.}$$

Check using area: set $f = 1$.

$$\begin{aligned} A &= \int_0^1 y\,\mathrm{d}y + \int_1^2 (2-y)\,\mathrm{d}y \\ &= \left[\frac{y^2}{2}\right]_0^1 + \left[2y - \frac{y^2}{2}\right]_1^2 = \frac{1}{2} + \frac{1}{2} = \boxed{1}. \end{aligned}$$

This agrees with the triangle formula $\frac{1}{2} \times 2 \times 1 = 1$.

Reversing order does not mean just swapping $\mathrm{d}x$ and $\mathrm{d}y$. Rebuild all bounds from the *same region*. Split at the outer-variable value where the boundary rule changes.

2. Reverse the order to make integration easier

$$I = \int_0^1 \int_x^1 e^{y^2}\,\mathrm{d}y\,\mathrm{d}x.$$

The inner antiderivative $\int e^{y^2}\,\mathrm{d}y$ is not elementary. Instead, use

$$0 \le x \le 1, \quad x \le y \le 1 \quad \Longleftrightarrow \quad 0 \le y \le 1, \quad 0 \le x \le y.$$

For fixed y, e^{y^2} is constant with respect to x. Therefore

$$\begin{aligned} I &= \int_0^1 \int_0^y e^{y^2}\,\mathrm{d}x\,\mathrm{d}y = \int_0^1 [xe^{y^2}]_0^y\,\mathrm{d}y = \int_0^1 ye^{y^2}\,\mathrm{d}y \\ &= \frac{1}{2}\int_0^1 e^u\,\mathrm{d}u \quad (u = y^2,\ \ \mathrm{d}u = 2y\,\mathrm{d}y) \\ &= \frac{1}{2}[e^u]_0^1 = \boxed{\frac{e-1}{2}}. \end{aligned}$$

The useful factor y comes from the horizontal slice width. The function e^{y^2} itself has not changed.

3. Area and volume over the same base

A material layer has the triangular base from Question 1. Coordinates are in centimetres and its thickness is $h(x, y) = 2 + x$ cm.

$$A = \iint_R 1\,\mathrm{d}A = 1 \text{ cm}^2.$$

For volume, replace the integrand 1 by the thickness. Since $0 \le x \le 1$, the thickness is positive everywhere.

$$\begin{aligned} V &= \int_0^1 \int_x^{2-x} (2 + x)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1 [(2 + x)y]_{y=x}^{y=2-x}\,\mathrm{d}x \\ &= \int_0^1 (2 + x)(2 - 2x)\,\mathrm{d}x \\ &= \int_0^1 (4 - 2x - 2x^2)\,\mathrm{d}x \\ &= \left[4x - x^2 - \frac{2x^3}{3}\right]_0^1 = \boxed{\frac{7}{3}\ \text{cm}^3}. \end{aligned}$$

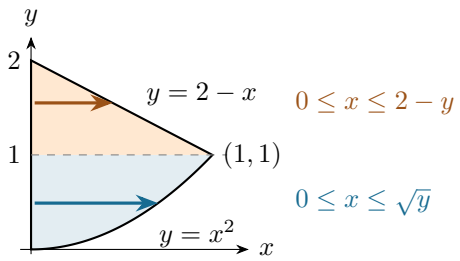
Check: the base area is 1 cm^2 and the thickness is between 2 and 3 cm. Therefore the volume must lie between 2 and 3 cm^3, as it does.

4. Try a curved boundary

Reverse the order in

$$J = \int_0^1 \int_{x^2}^{2-x} f(x,y)\,\mathrm{d}y\,\mathrm{d}x.$$

Before reading the solution, write the full y-range and the two right-hand boundaries.



Solution. The bounds describe $0 \le x \le 1$ and $x^2 \le y \le 2-x$. The curves meet where $(x+2)(x-1) = 0$; only $x = 1$ lies in this region, giving the split height $y = 1$. The complete y-range is $0 \le y \le 2$.

- For $0 \le y \le 1$, the slice runs from $x = 0$ to $x = \sqrt{y}$.
- For $1 \le y \le 2$, it runs from $x = 0$ to $x = 2 - y$.

The positive square root is required because the original region has $x \ge 0$.

$$\boxed{J = \int_0^1 \int_0^{\sqrt{y}} f(x,y)\,\mathrm{d}x\,\mathrm{d}y + \int_1^2 \int_0^{2-y} f(x,y)\,\mathrm{d}x\,\mathrm{d}y.}$$

For its area, set $f = 1$:

$$A = \int_0^1 \sqrt{y}\,\mathrm{d}y + \int_1^2 (2-y)\,\mathrm{d}y = \left[\frac{2}{3}y^{3/2}\right]_0^1 + \left[2y - \frac{y^2}{2}\right]_1^2 = \boxed{\frac{7}{6}}.$$

Before submitting: sketch the region; label intersections; identify the entry and exit of a typical slice; split if the boundary formula changes; use 1 for area and the correct height for volume; include units where given.